Microelectronics Curriculum

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SEAP

SCIENCE AND ENGINEERING APPRENTICE PROGRAM

Projective Objective and Intern Contribution:
Microelectronics is a growing field in the United States and there is a shortage of professionals able to work for the field. This summer we were tasked to work with Purdue University's Engineering Education Department as we represent NSWC Crane for the Scalable Asymmetric Lifecycle Engagement (SCALE) K-12 Program. The SCALE Program focuses on integrating engineering into broader contexts including the arts of electronics to create a more diverse group of students.

We tested and developed microelectronics material for the project primarily to elementary and middle school students using the micro:bit microcontroller and paper circuit electronics. We used lesson plans and presentations to teach both programming and electrical concepts through hands-on activites.

I worked on communicating with the PhD Candidates from Purdue University and taught the course material, such as the theatre-based paper circuit design to the STEM camp. Integrating art and engineering, we hope to attract a more diverse set of future scientists and engineers.

1. I am most proud of the students' creation as they showed their newly gained skillset. The students utilized their new knowledge ranging from the micro:bit's analog input/output to the radio communic ation features.
2. The internship was valuable because it introduced me to engineering education that I would have been unable to experience otherwise.
3. I would advise for future cohorts to learn to communicate well. Talking with my intern peers was quite an important skill I learned with the project and helped with camp coordination.

Results / Accomplishments / Next Steps:
Students often tend to dissociate engineering principles from other fields. Projects, such as creating a piano through tinfoil strips and printer paper, help illustrate the simplicity of engineering within art.

One of the deliverables was to create curriculum for a two-day STEM camp, themed "Art of Electronics". We demonstrated that these future engineers are better engaged through art from the results in the final project. The Navy is able to tap into these students to develop a sustainable workforce.